

# **Sri Lanka Institute of Information Technology**

## **Faculty of Computing**



### **EduFrame**

#### **Web-Based Video Browsing System**

*Web-Based Platform for Managing and Delivering Educational Video Content*

**Group – Y2-S1-MLB-B4G1-06**

**Year 2, Semester 1 – 2026**

**Lab Sheet 06 – Software Testing**

| IT Number         | Name                          |
|-------------------|-------------------------------|
| <b>IT25101841</b> | <b>De Silva L. C. A.</b>      |
| <b>IT25102695</b> | <b>Munasinghe M. H. V. S.</b> |
| <b>IT25103702</b> | <b>Bandara W. G. M. D.</b>    |
| <b>IT25101767</b> | <b>Withana T. T.</b>          |
| <b>IT25103673</b> | <b>Wickramathunga P. P.</b>   |
| <b>IT25100859</b> | <b>Adikari A. M. T. D.</b>    |

## 01: De Silva L. C. A. (IT25101841) - Announcement & Event System

| ID       | Title                          | Description                                                                                      | Preconditions                                                               | Steps                                                                                                                              | Data                                                                                | Expected                                                                    | Actual       | Status |
|----------|--------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|--------------|--------|
| TC-AN-01 | Create Valid Announcement      | Verify that an instructor can successfully publish a plain text announcement.                    | User is logged in as an instructor with active permissions.                 | 1. Navigate to 'Create Post'.<br>2. Select 'Announcement'.<br>3. Enter valid title and description.<br>4. Click 'Publish'.         | Title: 'Midterm Exam Notice'<br>Description: 'The exam will be held online.'        | Post saved in database, confirmation alert is shown.                        | As Expected, | Pass   |
| TC-AN-02 | Create Valid Academic Event    | Verify that an instructor can publish a live webinar event with calendar details.                | User is logged in as an instructor.                                         | 1. Navigate to 'Create Post'.<br>2. Select 'Event'.<br>3. Enter Title, Description, Date, Time, Venue Link.<br>4. Click 'Publish'. | Title: 'SE Revision Live'<br>Date: 2026-10-15<br>Time: 10:00 AM<br>Venue: Zoom Link | Event record created system routes control to notification engine.          | As Expected, | Pass   |
| TC-AN-03 | Schedule Conflict Validation   | Verify the system intercepts overlapping event schedules for a course to prevent double-booking. | An existing event is scheduled on the target date/time for that course.     | 1. Click 'Create Event'.<br>2. Enter matching date and time slots as an old event.<br>3. Click 'Publish'.                          | Date: 2026-10-15<br>Time: 10:00 AM<br>Course ID: SE2030                             | Form blocks submission and displays a 'Schedule Conflict Error'.            | As Expected  | Pass   |
| TC-AN-04 | Empty Mandatory Form Fields    | Verify validation triggers errors when trying to post an empty announcement.                     | User is on the Announcement Creation layout screen.                         | 1. Leave Title field completely blank.<br>2. Fill description payload.<br>3. Click 'Publish' button.                               | Title: "<br>Description: 'Valid content'                                            | System displays 'Title field is mandatory' input boundary warning.          | As Expected  | Pass   |
| TC-AN-05 | Real-time Dashboard Rendering  | Verify that a newly published event displays instantly on the student dashboard feed.            | An event is successfully stored in the system database.                     | 1. Log in as an enrolled Student.<br>2. Load the main user landing dashboard page.<br>3. Check the feed widget list.               | Student ID: IT25103702<br>Course ID: SE2030                                         | The newly created event appears on top of the dashboard feed.               | As Expected  | Pass   |
| TC-AN-06 | System Notification Dispatch   | Verify the system notification engine broadcasts alerts concurrently when a post goes live.      | A post completes database insertion validation without conflict.            | 1. Instructor clicks 'Publish'.<br>2. Check the system's background execution logs for the notification worker.                    | Post ID: P-991<br>Target Users: All enrolled students                               | Notification Service logs show parallel broadcast tasks fired successfully. | As Expected  | Pass   |
| TC-AN-07 | Post Modification Update       | Verify that an instructor can dynamically alter the venue or timing of an active event.          | The targeted event has been previously published and is live.               | 1. Open published posts log.<br>2. Select 'Update Details'.<br>3. Edit venue link fields.<br>4. Click 'Save Changes'.              | Updated Venue: 'Teams Link'<br>Post ID: P-991                                       | Changes are committed to DB, and updated info displays globally.            | As Expected  | Pass   |
| TC-AN-08 | Automated Expired Cleanup Hook | Verify that past or completed events are automatically cleaned from student dashboards.          | The event's calendar end-time has passed relative to the current timestamp. | 1. Wait for system cron job scheduler interval loop to trigger.<br>2. Inspect the live student dashboard.                          | Current Time: 2026-10-16<br>Event Time: 2026-10-15                                  | Expired event is filtered out and no longer renders on active feeds.        | As Expected  | Pass   |
| TC-AN-09 | Unauthorized Posting Access    | Verify that an unauthenticated guest user cannot bypass endpoints to create a notice post.       | User session is not logged in or holds a guest visitor token status.        | 1. Try to directly hit backend route /api/announcements/create via API client tools.                                               | HTTP POST<br>Payload: {title: 'Hacked'}                                             | Server returns HTTP 401 Unauthorized status code.                           | As Expected  | Pass   |
| TC-AN-10 | Immediate Content Deletion     | Verify that deleting a cancelled announcement removes it immediately from all interfaces.        | The announcement is currently visible to students on the platform.          | 1. Log in as Admin.<br>2. Select active post.<br>3. Click 'Remove Post'.<br>4. Re-verify the student view dashboard.               | Post ID: P-995                                                                      | Announcement is deleted from the DB and disappears from the dashboard.      | As Expected  | Pass   |

## 02: Munasinghe M. H. V. S. (IT25102695) - Ticketing System

| ID       | Title                           | Description                                                                               | Preconditions                                                     | Steps                                                                                                                      | Data                                                                      | Expected                                                                    | Actual      | Status |
|----------|---------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------|-------------|--------|
| TC-TK-01 | Submit Ticket Form              | Verify that a student can successfully submit a technical support ticket.                 | Student is authenticated and logged into the platform dashboard.  | 1. Navigate to 'Support' page.<br>2. Click 'Submit New Ticket'.<br>3. Enter Subject and Description.<br>4. Click 'Submit'. | Subject: 'Video Playback Error'<br>Description: 'Stuck loading at 1080p.' | Ticket recorded with status 'Open' and a unique Reference ID generated.     | As Expected | Pass   |
| TC-TK-02 | Attach Screenshot File          | Verify the ticket submission form accepts valid image attachments for context.            | User is filling out the ticket submission form.                   | 1. Click 'Attach File'.<br>2. Select a valid .png file.<br>3. Click 'Submit Ticket Form'.                                  | File: error_log.png (250 KB)                                              | System uploads file successfully and match it with the Ticket ID.           | As Expected | Pass   |
| TC-TK-03 | Block Oversized Attachments     | Verify the ticket system rejects attachment files that exceed the size constraints.       | User is filling out the ticket submission form.                   | 1. Click 'Attach File'.<br>2. Upload a file exceeding 5MB.<br>3. Attempt to click 'Submit'.                                | File: large_video.mp4 (12 MB)                                             | Form rejects submission and shows error message 'File size limit exceeded'. | As Expected | Pass   |
| TC-TK-04 | Admin Queue Inspection          | Verify that platform admins can view incoming tickets categorized by status.              | Admin user is logged in with administrative roles.                | 1. Access the 'Admin Portal'.<br>2. Click 'Ticket Resolution Queue'.<br>3. Apply status filter 'Open'.                     | Filter Criteria: Status = 'Open'                                          | Queue renders a clear list of all unresolved tickets submitted by users.    | As Expected | Pass   |
| TC-TK-05 | Update Status to In Progress    | Verify that an admin opening a ticket changes its tracking state automatically.           | Admin selects an 'Open' ticket from the processing list.          | 1. Open ticket details page.<br>2. Click 'Investigate/Process'.<br>3. Save updated metrics.                                | Ticket Reference ID: #TK-8821                                             | Ticket status updates from 'Open' to 'In Progress' in the database matrix.  | As Expected | Pass   |
| TC-TK-06 | Post Admin Response Text        | Verify that admins can add contextual log messages or updates to an active ticket thread. | Admin is inside a ticket details page.                            | 1. Type message text into response box.<br>2. Click 'Post Reply'.                                                          | Response: 'We are clearing our CDN cache.'                                | Response text appended to the ticket view history chronologically.          | As Expected | Pass   |
| TC-TK-07 | Mark Ticket as Resolved         | Verify that closing a ticket changes its state and sends confirmation logs.               | Ticket issue has been fixed by administration staff.              | 1. Click 'Mark Resolved'.<br>2. Input the final solution writeup.<br>3. Submit the tracking update.                        | Final Note: 'CDN cache successfully reset.'                               | Ticket state flags as 'Resolved'; alert notification dispatched to user.    | As Expected | Pass   |
| TC-TK-08 | Duplicate Ticket Interception   | Verify that system flags or handles matching tickets sent concurrently by the same user.  | An identical ticket is already recorded in the system state.      | 1. Click 'Submit New Ticket'.<br>2. Input duplicate subject and payload details.<br>3. Click 'Submit'.                     | Subject: 'Video Playback Error'                                           | System flags duplicate status and logs the new submission as 'Closed'.      | As Expected | Pass   |
| TC-TK-09 | User Track Progress View        | Verify that students can view the history logs of their historical support tickets.       | Student has submitted tickets previously on the platform account. | 1. Navigate to 'Support Dashboard'.<br>2. Open 'My Tickets History Log'.                                                   | User Session: IT25102695                                                  | Renders historical list displaying matching Reference IDs and states.       | As Expected | Pass   |
| TC-TK-10 | System Performance Notification | Verify the alert engine pushes a dashboard flash message when an admin solves a ticket.   | An active ticket has its state flag changed to 'Resolved'.        | 1. Admin marks ticket resolved.<br>2. Log in as target student user account.                                               | Ticket Reference ID: #TK-8821                                             | System displays a visual notification alert showing resolution message.     | As Expected | Pass   |

### 03: Bandara W. G. M. D. (IT25103702) - Advertisement Promotions

| ID       | Title                          | Description                                                                              | Preconditions                                                                  | Steps                                                                                                                                  | Data                                                                             | Expected                                                                     | Actual      | Status |
|----------|--------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------|-------------|--------|
| TC-AD-01 | Create Advertisement Banner    | Verify that a Marketing Executive can successfully create a new ad slider banner.        | User is authenticated with marketing team privileges.                          | 1. Navigate to 'Promotion Manager'.<br>2. Select 'Create Advertisement'.<br>3. Input Title, MediaType, and Budget.<br>4. Click 'Save'. | Title: 'Special Cloud Comp Course'<br>MediaType: 'Banner Image'<br>Budget: 50000 | Record created in database with a baseline status of 'Scheduled'.            | As Expected | Pass   |
| TC-AD-02 | Upload Campaign Creative Image | Verify the advertisement form accepts standard graphical aspect ratio images.            | User is filling out the advertisement creation form interface.                 | 1. Click 'Upload Media File'.<br>2. Attach a valid .jpeg image asset.<br>3. Submit the promotional form data.                          | File: promo_banner.jpeg (800 KB)                                                 | Asset uploaded to image storage buckets and linked to ad ID entity.          | As Expected | Pass   |
| TC-AD-03 | Invalidate Empty Fields        | Verify that leaving required campaign text blank triggers form submission blocks.        | User is on the promotional management builder panel.                           | 1. Input Title information.<br>2. Leave Budget fields blank.<br>3. Attempt to save form data.                                          | Title: 'Big Discount'<br>Budget: "                                               | System displays form layout alert stating, 'Required ad fields empty'.       | As Expected | Pass   |
| TC-AD-04 | Dynamic Homepage Render        | Verify that active advertisements render dynamically inside the homepage viewport.       | Current date falls within the configured display timeline schedules.           | 1. Open platform main page url as an unauthenticated Visitor.<br>2. Observe top slider banner cards.                                   | Current Date: 2026-09-28<br>Ad Schedule: 2026-09-01 to 2026-10-01                | The targeted campaign banner shows inside the homepage promotional view.     | As Expected | Pass   |
| TC-AD-05 | Schedule Timeline Invalidation | Verify that campaigns do not display if their active display timeline is expired.        | System current timestamp is greater than campaign display timeline parameters. | 1. Load the main homepage listing feed.<br>2. Inspect slider index elements.                                                           | Current Date: 2026-09-28<br>Ad Expiry: 2026-09-25                                | The expired promotional card is automatically withheld from user feeds.      | As Expected | Pass   |
| TC-AD-06 | Video Streaming Ad Intercept   | Verify system queries promotional engines before starting educational video streaming.   | A visitor clicks on a public educational video thumbnail item card.            | 1. Select video lesson panel.<br>2. Check for commercial advertisement roll playback.                                                  | Video ID: V-1022<br>User: Visitor                                                | Video player pauses core lesson and fires a mandatory banner pre-roll block. | As Expected | Pass   |
| TC-AD-07 | Bypass Non-Ad Video Paths      | Verify that video playback continues uninterrupted if no ad slots match current context. | No active campaigns are scheduled in the promotional lookup tables.            | 1. Click on educational video page template.<br>2. Observe immediate player events.                                                    | Active Campaigns Matrix: Empty                                                   | Player bypasses ad roll states entirely and plays video stream directly.     | As Expected | Pass   |
| TC-AD-08 | Track Ad Interaction Click     | Verify the telemetry service records analytics data when users click campaign items.     | User clicks on a live displayed homepage promotion banner slider.              | 1. Click active homepage banner link card.<br>2. Check background analytical tracking logs.                                            | Campaign ID: #AD-501                                                             | Telemetry system registers an interaction entry log to calculate ROI.        | As Expected | Pass   |
| TC-AD-09 | Update Promotional Schedule    | Verify that marketing teams can modify display dates for an existing ad.                 | Target advertisement asset record is stored in database.                       | 1. Select active campaign log.<br>2. Edit display dates fields.<br>3. Click 'Save Modifications'.                                      | New End Date: 2026-11-01                                                         | System updates record values and re-validate visibility loops.               | As Expected | Pass   |
| TC-AD-10 | Immediate Campaign Deletion    | Verify deleting a campaign immediately purges the banner asset from user viewports.      | An advertisement banner card is currently rendering on visitor pages.          | 1. Admin selects campaign entity.<br>2. Click 'Force Delete Archive'.<br>3. Re-verify the visitor home view.                           | Campaign ID: #AD-501                                                             | Banner is deleted from database and disappears instantly from homepage.      | As Expected | Pass   |

## 04: Withana T. T. (IT25101767) - Course & Curriculum Management

| ID       | Title                           | Description                                                                             | Preconditions                                                                   | Steps                                                                                                                     | Data                                                           | Expected                                                                         | Actual      | Status |
|----------|---------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------------------------|-------------|--------|
| TC-CR-01 | Create Syllabus Module Entity   | Verify that an Instructor can successfully create a new structured course card.         | Instructor account is validated and logged in securely.                         | 1. Open 'Course Management Dashboard'.<br>2. Click 'Create Course'.<br>3. Enter Title, Category, Description.<br>4. Save. | Title: 'Advanced SE Frameworks'<br>Category: 'IT/Software Eng' | Course object initialized and stored inside system catalogue schema tables.      | As Expected | Pass   |
| TC-CR-02 | Add Curriculum Structural Units | Verify that instructors can subdivide a course framework into multi-tier chapters.      | Base course entity has been instantiated inside databases.                      | 1. Open Course workspace panel.<br>2. Click 'Add Module/Chapter'.<br>3. Type summary titles.<br>4. Click 'Append Module'. | Chapter Title: '01: Architecture Basics'                       | Chapter link matrix successfully attached as a sub-child element of course.      | As Expected | Pass   |
| TC-CR-03 | Course Catalogue Browsing       | Verify that visitors can browse full structured course catalogues by category filters.  | Course records are verified and flagged as 'Published' inside system lookups.   | 1. Navigate to public platform landing link.<br>2. Click 'Browse by Category Filters'.<br>3. Select 'IT/Software Eng'.    | Filter Category: 'IT/Software Eng'                             | Catalogue indexes load displaying all matches matching category keys.            | As Expected | Pass   |
| TC-CR-04 | Prevent Double Enrolment Path   | Verify that the system blocks multiple active enrollment rows for the same student.     | Student account is already actively registered inside the course roster matrix. | 1. Open enrolled course details page.<br>2. Attempt to trigger the 'Enroll' endpoint again.                               | User ID: IT25101767<br>Course ID: #CS-402                      | System blocks operation, showing notification 'Already enrolled in course'.      | As Expected | Pass   |
| TC-CR-05 | Successful Student Enrolment    | Verify that clicking enroll appends student profile variables to the roster fields.     | Student is authenticated and holds no historical links with the target course.  | 1. Click 'Enroll in This Course'.<br>2. Validate enrollment response codes.                                               | Course ID: #CS-405<br>Student Session ID: Active               | Roster logs insert relationship; course appended to user dashboard list.         | As Expected | Pass   |
| TC-CR-06 | Block Malformed Metadata Forms  | Verify text fields assert validation boundaries when values contain illegal lengths.    | Instructor is typing summary notes into the course builder panel.               | 1. Input short title text.<br>2. Fill long description fields exceeding 5000 strings.<br>3. Click 'Save'.                 | Description Length: 6000 chars                                 | Core validation loops intercept payload and reject database commit steps.        | As Expected | Pass   |
| TC-CR-07 | Update Course Details           | Verify that instructors can alter description and context details post-publication.     | Targeted course card is live and accessible on catalogue interfaces.            | 1. Select owned course entry log.<br>2. Click 'Modify Course Matrix'.<br>3. Rewrite summary text fields.<br>4. Save.      | New Title: 'Intro to UML Modeling'                             | Database updates the targeted columns; layout updates display values.            | As Expected | Pass   |
| TC-CR-08 | Archive Completed Course Track  | Verify that archiving a course changes visibility logic without breaking student logs.  | Target course is old or no longer accepting active enrollments.                 | 1. Click 'Archive Course Track'.<br>2. Verify public catalog status listing states.                                       | Course ID: #CS-402                                             | Catalogue removes public display cards; existing students retain history access. | As Expected | Pass   |
| TC-CR-09 | Enrolment Count Telemetry       | Verify the aggregation counter tracking active students increments automatically.       | A new verified student profile registers an enrollment action.                  | 1. Student executes 'Enroll'.<br>2. Admin inspects dashboard analytics figures.                                           | Course ID: #CS-405                                             | Total student tracking metric increments by +1 inside telemetry rows.            | As Expected | Pass   |
| TC-CR-10 | Cross-Module Module Linkage     | Verify that course modules allow secondary items like video links or quizzes to attach. | Course structural chapters are successfully mapped inside tables.               | 1. Select specific module chapter node.<br>2. Assign a video resource object ID to it.<br>3. Confirm association.         | Video Object ID: V-90321                                       | Structural database maps child dependencies without throwing key errors.         | As Expected | Pass   |

## 05: Wickramathunga P. P. (IT25103673) - Video Management

| ID       | Title                           | Description                                                                          | Preconditions                                                                   | Steps                                                                                                   | Data                                                          | Expected                                                                           | Actual      | Status |
|----------|---------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------------------|-------------|--------|
| TC-VD-01 | Upload Valid Media Container    | Verify that an instructor can successfully upload a standard educational video file. | Instructor is logged in and working inside an assigned course workspace.        | 1. Open 'Upload Lesson Form'.<br>2. Select video file asset.<br>3. Fill title parameters.<br>4. Upload. | File: lecture_01.mp4<br>Size: 150 MB<br>Title: 'UML Overview' | Binary file pipes to cloud bucket targets; database records state metadata.        | As Expected | Pass   |
| TC-VD-02 | Unsupported Container Block     | Verify that system validation filters out dangerous or unoptimized extensions.       | User is selecting local file objects inside file picker dialogues.              | 1. Select a file with a non-video format extension.<br>2. Attempt to upload the asset.                  | File: hack_script.exe<br>Size: 5 MB                           | Upload engine blocks execution, outputting error 'Unsupported file format'.        | As Expected | Pass   |
| TC-VD-03 | Oversized Media Rejection       | Verify system boundary scripts drop video upload requests exceeding max sizes.       | Instructor opens the multimedia ingestion interface.                            | 1. Select heavy video asset file.<br>2. Click 'Execute Upload Stream'.                                  | File Size: 1.8 GB<br>Max Allowed: 1.0 GB                      | Gateway closes connections, showing error text 'File size limits exceeded'.        | As Expected | Pass   |
| TC-VD-04 | Duration Constraint Enforcement | Verify the analyzer tracks duration fields and rejects files exceeding limits.       | A video media upload request initiates stream processing pass.                  | 1. Upload a video file containing high playback duration timelines.                                     | Duration: 45 Minutes<br>Max Limit: 30 Minutes                 | Processing engine drops transaction; notifies user 'Video duration exceeds limit'. | As Expected | Pass   |
| TC-VD-05 | Form Progress Telemetry Loop    | Verify the interface layout accurately presents progressive stream upload states.    | Data transport tunnels are running binary streams to hosting clouds.            | 1. Observe interface status bar animations during active upload states.                                 | Stream Rate: 10 Mbps                                          | Visual UI updates rendering active completion scale numbers (e.g., 45% -> 46%).    | As Expected | Pass   |
| TC-VD-06 | Route to Moderation Queue       | Verify that successfully ingested videos sit inside a quarantine holding area.       | Ingestion engine completes pipeline writes without file structure errors.       | 1. Log file transfers values.<br>2. Open administrative review dashboards panels.                       | Video Reference: #VD-1092                                     | Video state flags as 'Under Review'; held back from student dashboards.            | As Expected | Pass   |
| TC-VD-07 | Automated Banned Content Stop   | Verify keyword validation filters cross-reference titles with platform blocklists.   | Content blocklists contain specific prohibited strings inside metadata lookups. | 1. Input blacklisted keyword in title text.<br>2. Click 'Publish'.                                      | Title: 'Illegal Streaming Hack'                               | Moderation engine flags record, shifting item status to 'Quarantined'.             | As Expected | Pass   |
| TC-VD-08 | Smooth HLS Playback Ingest      | Verify that student profiles can invoke HTML5 playback stream pipelines easily.      | Video status has been approved and moved into global database catalogues.       | 1. Log in as Student.<br>2. Click video item link preview cards.<br>3. Check player canvas states.      | Video ID: #VD-1092                                            | Canvas updates frame renders; view aggregation telemetry tracks streaming inputs.  | As Expected | Pass   |
| TC-VD-09 | Update Video Metadata           | Verify creators can alter presentation thumbnails or summaries without re-uploading. | Target multimedia file row is verified inside database indexes.                 | 1. Access owned asset workspace.<br>2. Upload new thumbnail image file.<br>3. Click 'Refresh Details'.  | File: new_thumb.png                                           | Database overwrites paths configuration records; preview windows update.           | As Expected | Pass   |
| TC-VD-10 | Immediate Video Deletion        | Verify permanent deletion purges entries from both tables and file systems.          | Content creator chooses to remove outdated lecture records from index grids.    | 1. Click 'Delete Media Content'.<br>2. Confirm database check prompts.                                  | Video ID: #VD-1092                                            | System clears database index columns and unlinks binary files from cloud storage.  | As Expected | Pass   |

## 06: Adikari A. M. T. D. (IT25100859) - Quiz & Assessment Subsystem

| ID | Title | Description | Preconditions | Steps | Data | Expected | Actual | Status |
|----|-------|-------------|---------------|-------|------|----------|--------|--------|
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|----------|----------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------|-------------|------|
| TC-QZ-01 | <b>Initialize Timed Quiz Form</b>      | Verify that an instructor can create a timed evaluation card inside a module.                    | Instructor user is logged in with active subsystem management rights.       | 1. Select target course module.<br>2. Click 'Create Quiz'.<br>3. Input Title, Duration, PassMark.                       | Title: 'UML Architecture Quiz'<br>Duration: 15 Mins<br>PassMark: 70 | Quiz configuration header committed successfully to database schema lookups.     | As Expected | Pass |
| TC-QZ-02 | <b>Populate Question Bank Row</b>      | Verify form allows injection of multiple-choice options tied to a specific quiz.                 | Quiz configuration container header row is successfully initialized.        | 1. Click 'Add Question'.<br>2. Type query text fields.<br>3. Input four choices; select correct answer box.<br>4. Save. | Question: 'What does BCE stand for?'<br>Options: [A, B, C, D]       | Question row inserted into bank array schema; answer key map verified.           | As Expected | Pass |
| TC-QZ-03 | <b>Enforce Invalid Duration Checks</b> | Verify system checks block attempts to register negative or zero timer metrics.                  | User is building an evaluation quiz framework workspace.                    | 1. Input valid title notes.<br>2. Enter invalid configuration numbers into duration inputs.<br>3. Save.                 | Title: 'Quick Test'<br>Duration: -5 Minutes                         | System blocks saving operations, showing text alert 'Invalid duration entered'.  | As Expected | Pass |
| TC-QZ-04 | <b>Start Test Countdown Engine</b>     | Verify student workspace panel initializes countdown timers accurately on load.                  | Student profile holds active enrollment links and opens a quiz page.        | 1. Review instructions text cards.<br>2. Click 'Start Quiz'.<br>3. Observe top timer status bars.                       | Quiz ID: #QZ-441                                                    | Timer engine registers clock down sequences from base configurations (15:00).    | As Expected | Pass |
| TC-QZ-05 | <b>Form Entry State Preservation</b>   | Verify that clicking radio choice controls sets temporary user response arrays.                  | Evaluation test workspace is currently open and ticking down time metrics.  | 1. Click target radio button response item next to question text blocks.                                                | Selection: Choice [B]                                               | Active memory cache binds profile response rows to the selected question node.   | As Expected | Pass |
| TC-QZ-06 | <b>Enforce Expiry Auto-Submit</b>      | Verify the test engine locks interfaces and pushes payloads when the clock hits zero.            | Active countdown timer metrics decrement down to absolute zero boundaries.  | 1. Allow the active canvas countdown clock to reach 00:00 without clicking submit.                                      | Time Metric: 00:00                                                  | System disables canvas text inputs, captures state, and sends submission data.   | As Expected | Pass |
| TC-QZ-07 | <b>Automated Grading Calculation</b>   | Verify backend processors compare response values with keys to calculate accurate scores.        | Student clicks explicit submit button before expiration boundaries trigger. | 1. Click 'Finalize Submission'.<br>2. Verify response calculation matrix events.                                        | Student Answers: 8 Correct / 2 Wrong                                | Grading engines outputs performance values matching raw scores (e.g., 80%).      | As Expected | Pass |
| TC-QZ-08 | <b>Update Pass/Fail Status Log</b>     | Verify comparison checking updates passing status records inside student tables.                 | Evaluation score is algorithmically processed by grading engine scripts.    | 1. Check database evaluation metrics.<br>2. Compare score values with thresholds.                                       | Score: 80%<br>Threshold<br>PassMark: 70%                            | System updates status column parameters to 'Pass' inside profile dashboards.     | As Expected | Pass |
| TC-QZ-09 | <b>Trigger Certificate Generator</b>   | Verify passing status states execute background PDF creation engines successfully.               | Performance score metrics successfully bypass pass mark constraint scales.  | 1. Complete test with high score values.<br>2. Observe post-submission link screens.                                    | Status: Pass                                                        | PDF building microservice prints unique validation credentials and course tags.  | As Expected | Pass |
| TC-QZ-10 | <b>Block Failed Profile Rewards</b>    | Verify the certificate generation script is withheld if performance trends sit below pass marks. | Performance score output metrics sit beneath target threshold scale steps.  | 1. Submit quiz responses containing high error rates.<br>2. Check profile records paths.                                | Status: Fail<br>Score: 40%                                          | System locks reward downloads interfaces, showing instead 'Retake Test' options. | As Expected | Pass |